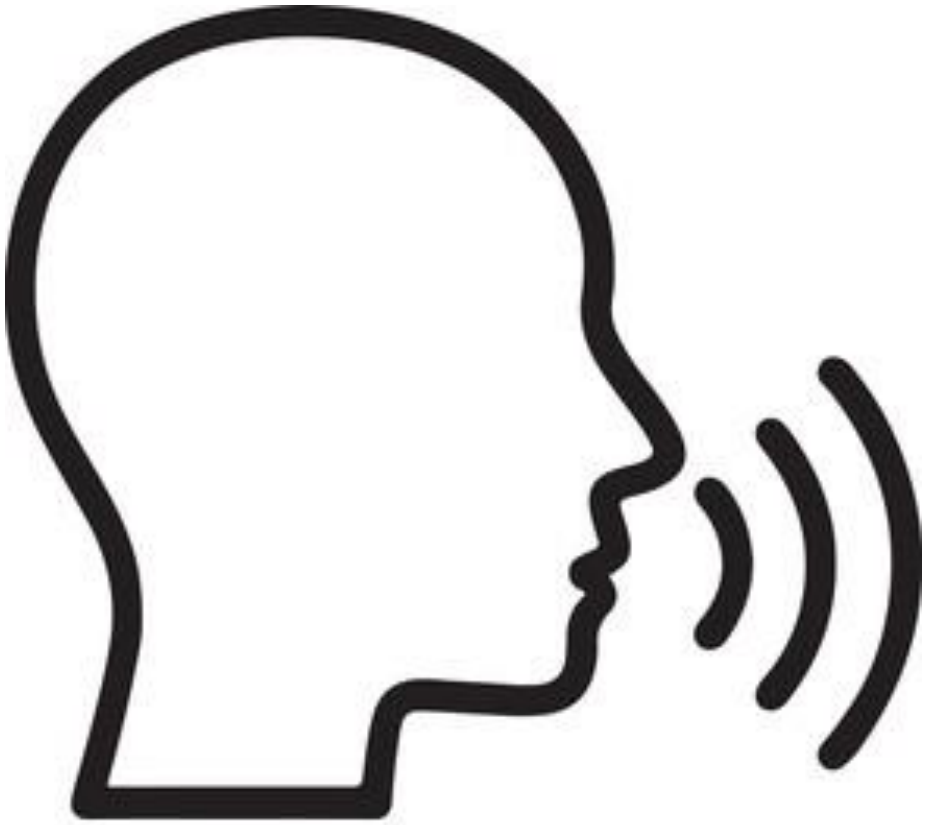
EmoTracker

Alex, Aaron, and Matt

Emotion Tracking?



- Emotion is expressed through face, voice, and words, simultaneously
- Most computer systems can only analyze one channel at a time, losing critical context
- Gaps in telehealth, online learning, and customer support are the direct cost
- Existing tools (Imentiv AI, iMotions, MorphCast) are expensive, privacy-invasive, or enterprise-only

EmoTracker: Real-Time Multi-Modal Emotion Recognition

An accessible, local, privacy-friendly alternative that fuses all three modalities

Five Emotions:
Happy · Sad · Angry · Fearful · Neutral

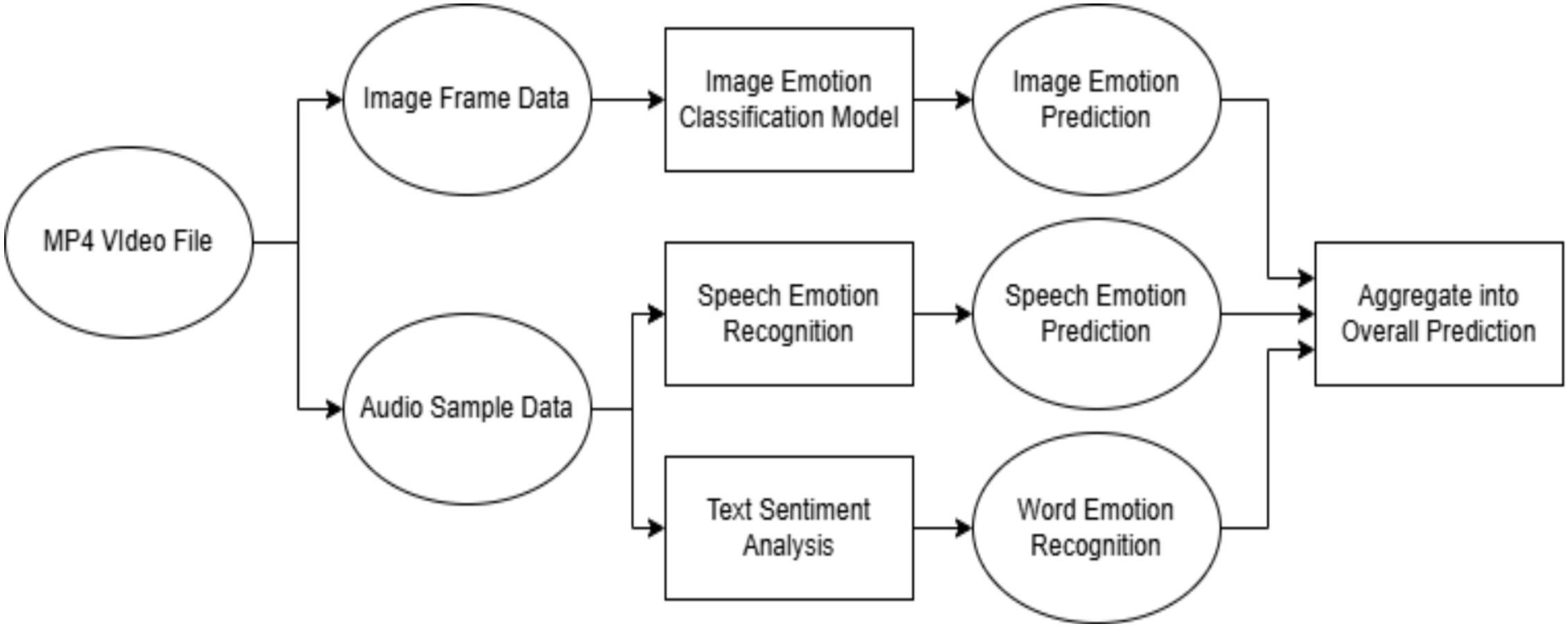
Related Work

Reference	Type	Approach	Strength	Limitation
Oheix — FER2013 (Kaggle)	FER	CNN on FER2013	Solid facial baseline	Single modality; limited generalization
Ozturk — VGG19/FER2013 (Kaggle)	FER	VGG19 deep CNN	Strong deep feature extraction	Still under-performs on real-world variation
Burnwal — SER (Kaggle)	SER	Crema-D/RAVDESS/SAVEE/TESS	Broad multi-dataset SER	Single modality; sensitive to noise
Zahara et al. — IEEE 2020	SER	Hybrid CNN + RNN	Captures temporal audio patterns	Lab conditions; limited diversity
Ucak — Twitter Emotion (Kaggle)	TER	CNN on tweet text	Effective short-form text classification	Struggles with sarcasm and ambiguity

System Architecture

Component	Technology
FER Model	OpenCV (Haar Cascade) + VGGNet CNN: Keras/TensorFlow
SER Model	Librosa (MFCC/chroma/mel) + BiLSTM + CNN: Keras
TER Model	HuggingFace Transformers + BERTweet (vinai/bertweet-base)
ASR	Automatic Speech Recognition: speech-to-text preprocessing for TER
Dashboard	Python Flask · JS · HTML/CSS · Google Fonts

System Architecture (2)



Data Preprocessing

Video / FER:

- Dataset: FER2013 (~31,000 images, with diverse emotions, and intensities)
- OpenCV + Haar Cascade Classifier for face detection and crop per frame
- Frames resized and normalized; image augmentation applied during training
- Class distribution was relatively equal but used class weights to eliminate any bias from unequal classes.

Audio / SER:

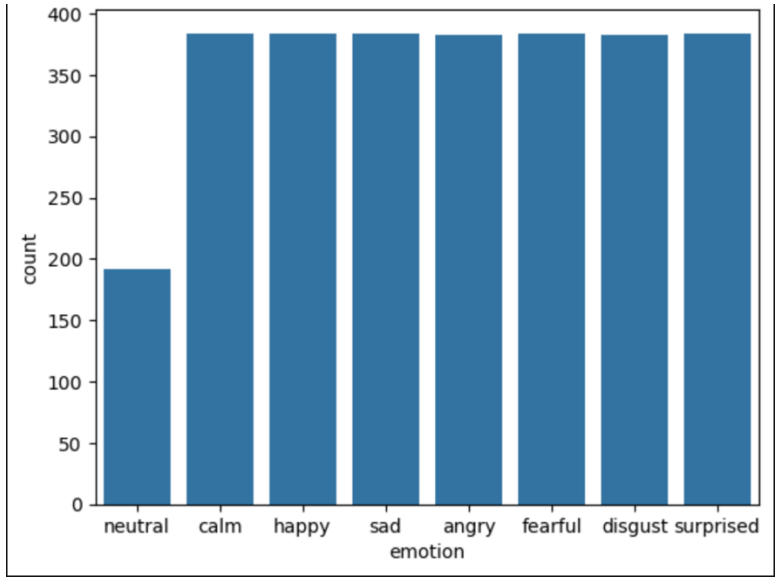
- Datasets: Crema-D, RAVDESS, compiled into one dataset (~15,000 samples, ~2,500 per class)
- "Surprise" dropped — insufficient data to support as a full class
- SAVEE (male-only) and TESS (female-only, high sample count) flagged for gender bias; considered for exclusion
- MFCCs, chroma, and mel-spectrogram features extracted via Librosa; audio normalized for volume and silence

Text / TER:

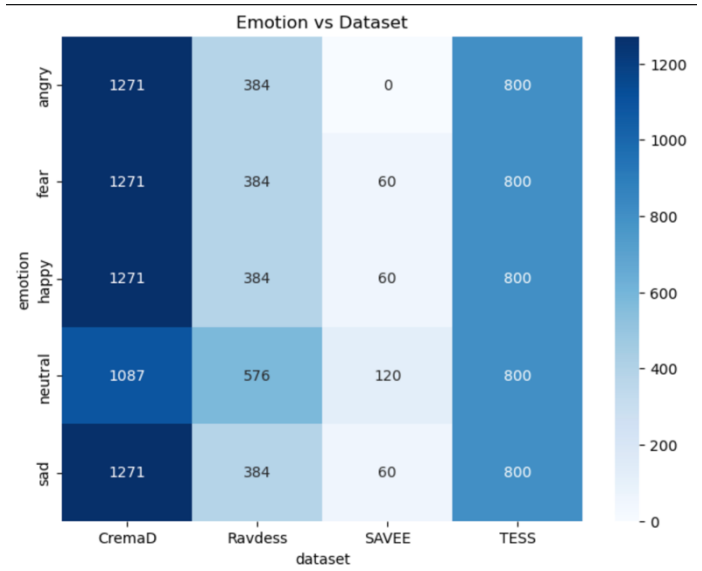
- Datasets: pashupatigupta (~40K tweets, 13 labels) + bhavikjikadara/emotions-dataset
- 13 labels remapped to 6 consolidated classes; "empty" label manually inspected and merged into Neutral
- Text cleaned: lowercased, removed @mentions, URLs, punctuation, whitespace
- Labels encoded with scikit-learn's LabelEncoder; 80/20 stratified train/test split
- Class imbalance (Happy: 13,112 vs. Angry: 1,433) handled via class_weight='balanced' — no downsampling (would have discarded ~31K rows)

EDA - Distributions

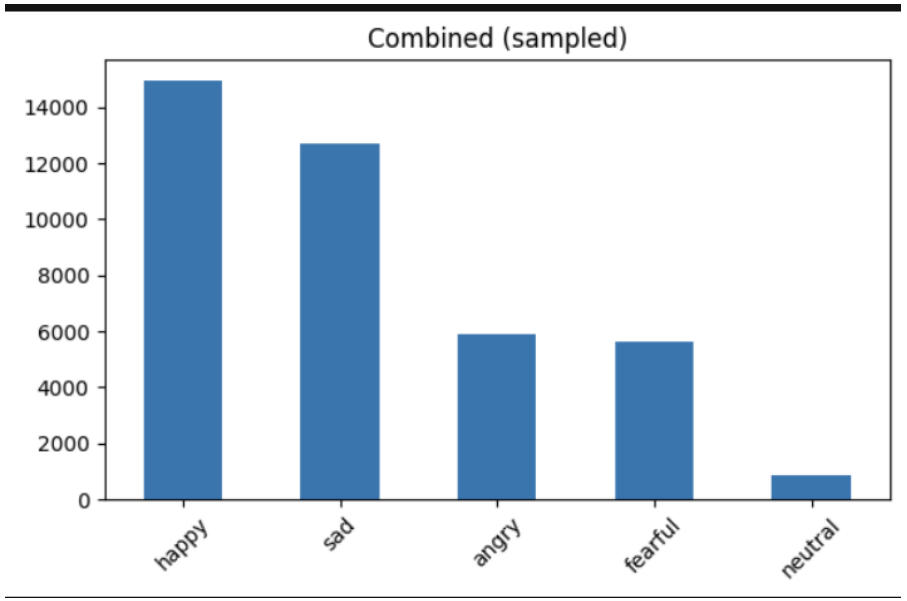
FER



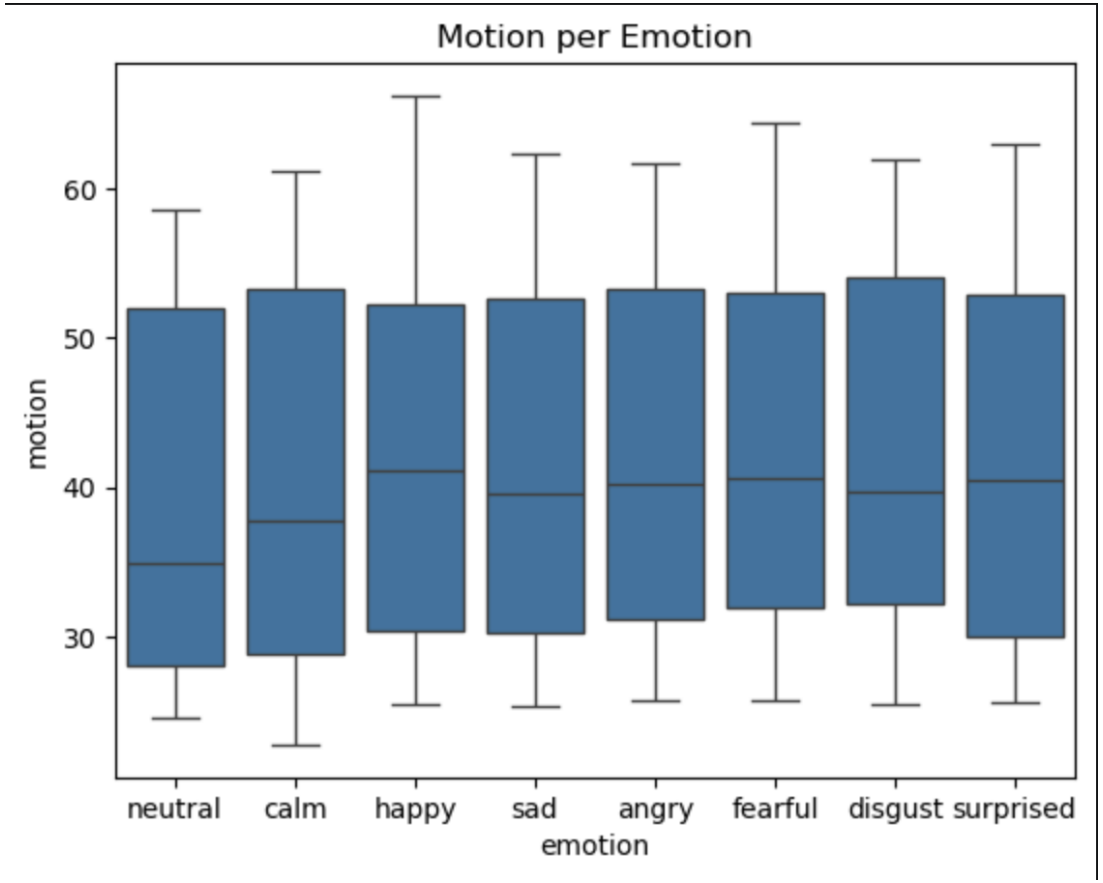
SER

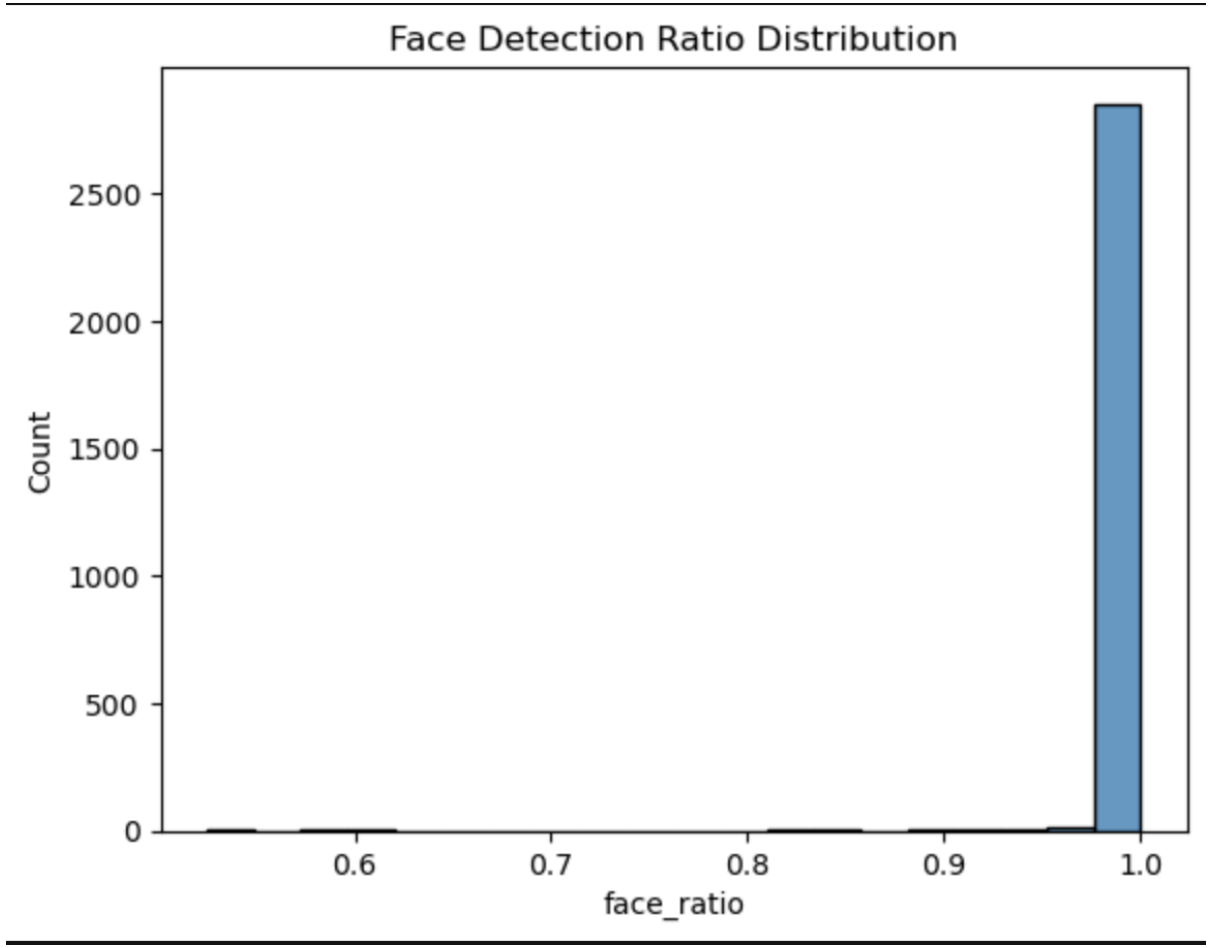


TER

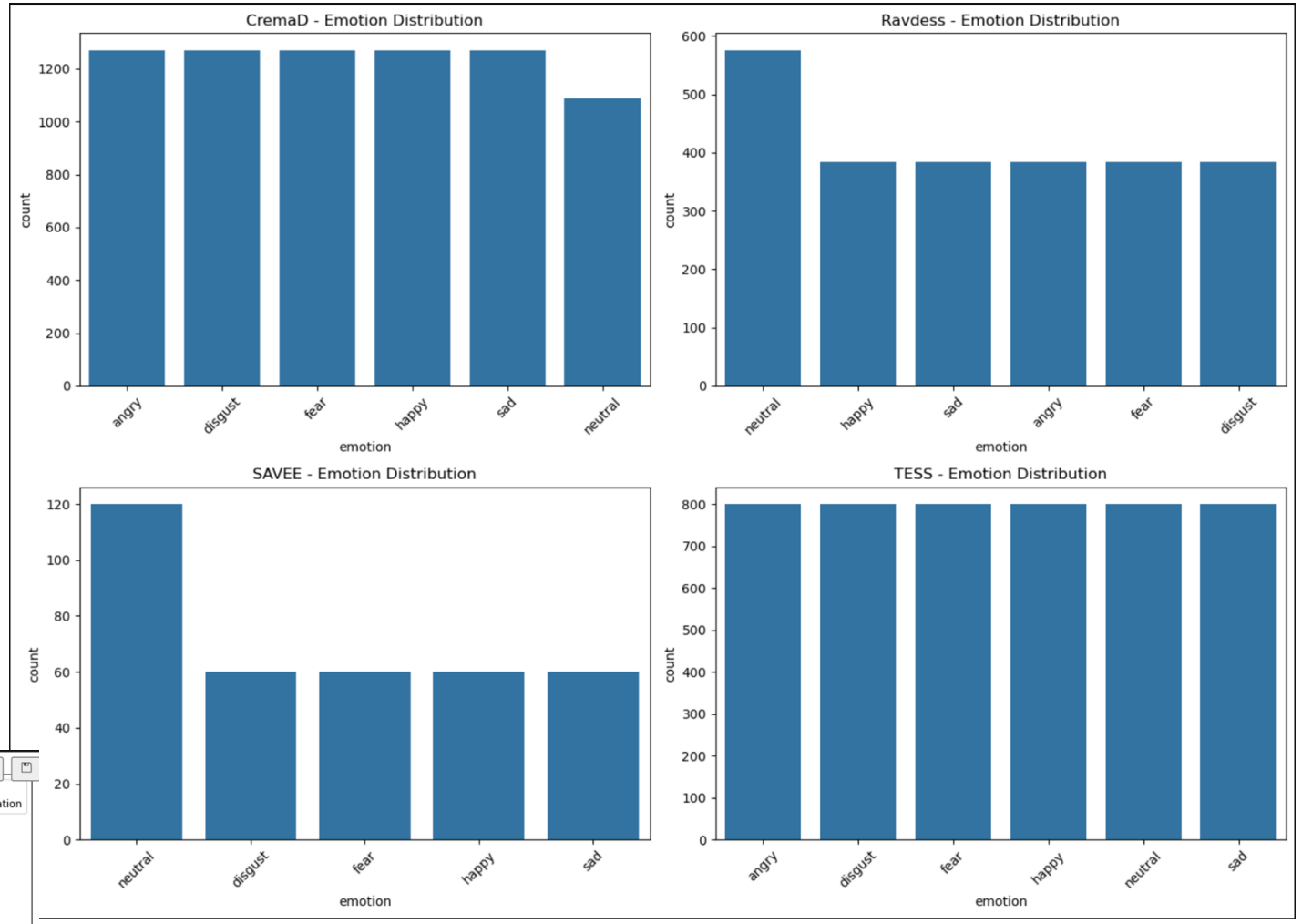


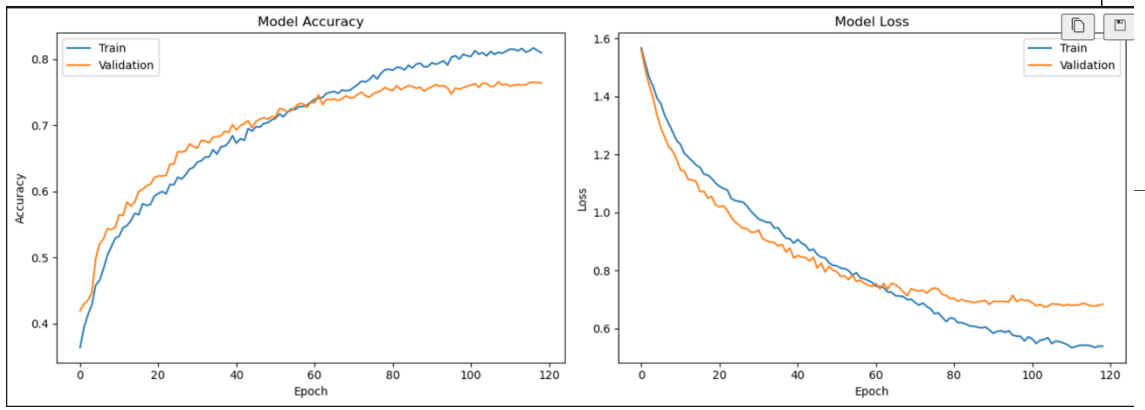
EDA — Video Data Analysis (RAVDESS / FER)



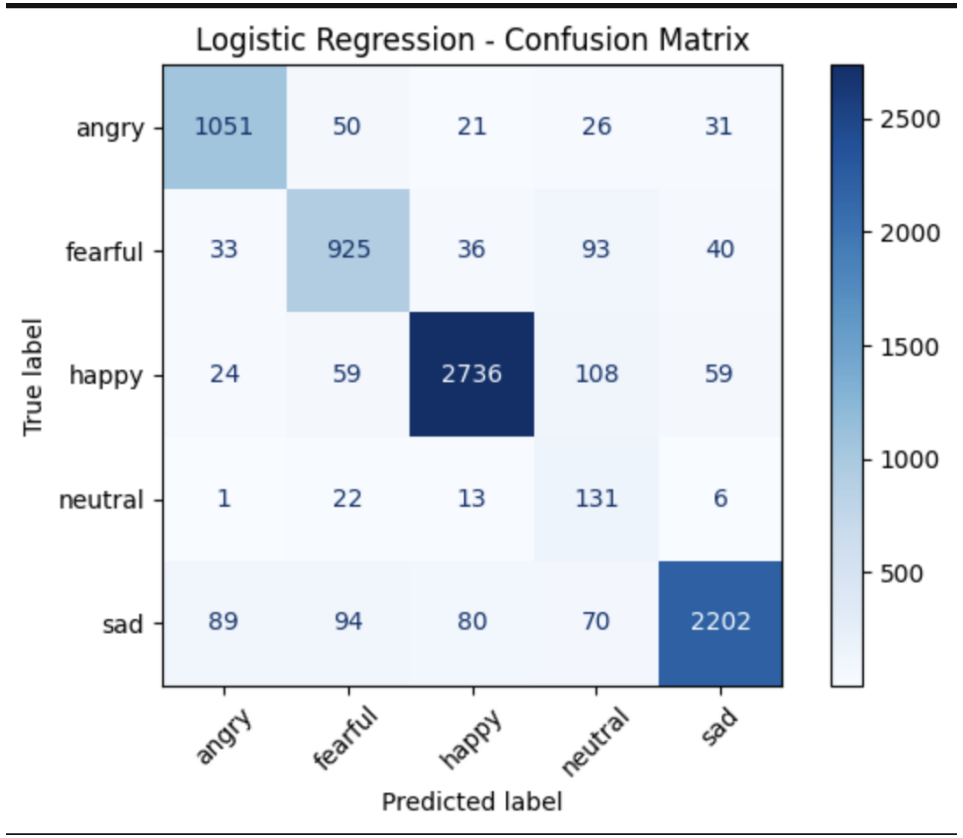


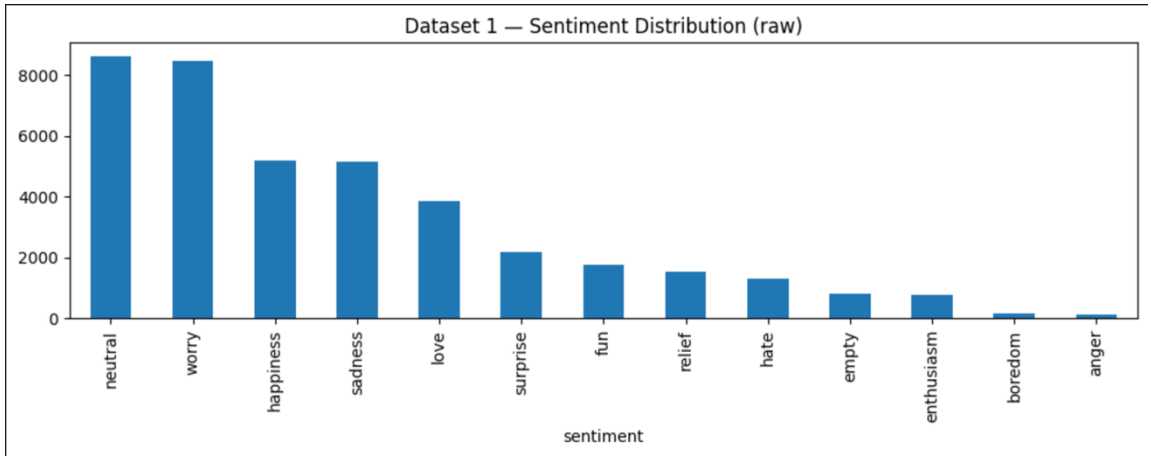
EDA — Audio Data Analysis (SER Datasets)

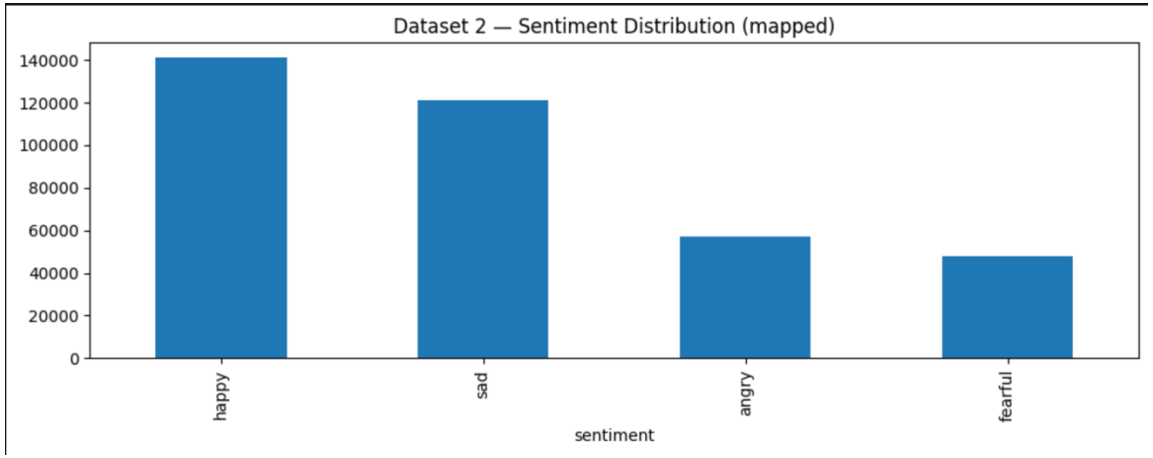




EDA — Text Sentiment Distribution (TER Dataset)





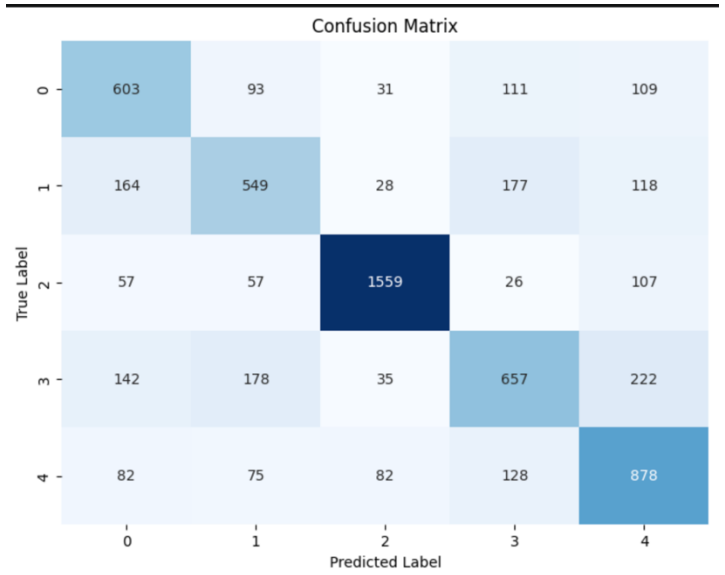


Model 1 — Facial Expression Recognition (FER)

- Face detection: OpenCV + Haar Cascade Classifier per video frame
- Architecture: VGGNet-based CNN (Keras/TensorFlow) — investigated after initial CNN+LSTM work on RAVDESS
- Training: FER2013

Classification Report:

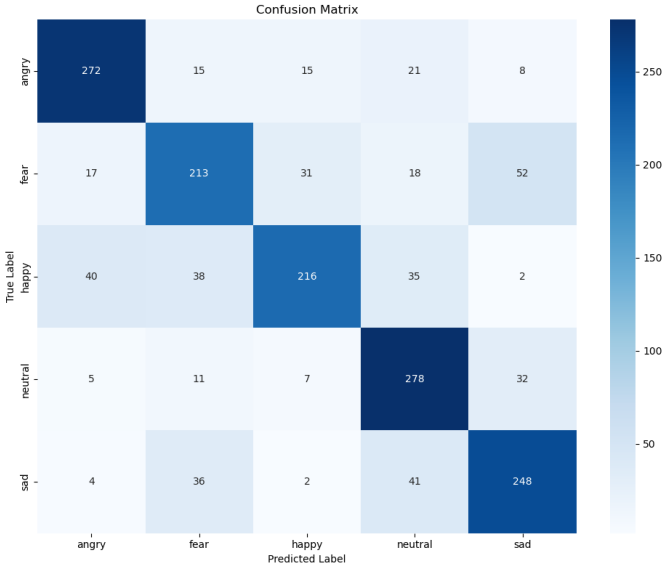
	precision	recall	f1-score	support
0	0.58	0.64	0.60	947
1	0.58	0.53	0.55	1036
2	0.90	0.86	0.88	1806
3	0.60	0.53	0.56	1234
4	0.61	0.71	0.66	1245
accuracy			0.68	6268
macro avg	0.65	0.65	0.65	6268
weighted avg	0.68	0.68	0.68	6268



Model 2 — Speech Emotion Recognition (SER)

- Feature extraction: Librosa — MFCCs, chroma, mel-spectrogram
- Architecture: BiLSTM + CNN (Keras) — BiLSTM captures temporal audio patterns; CNN finds spectral features
- Trained on compiled dataset: Crema-D + RAVDESS (~12,000 samples)
- Compared two architectures: BiLSTM+CNN vs. CNN+BiLSTM

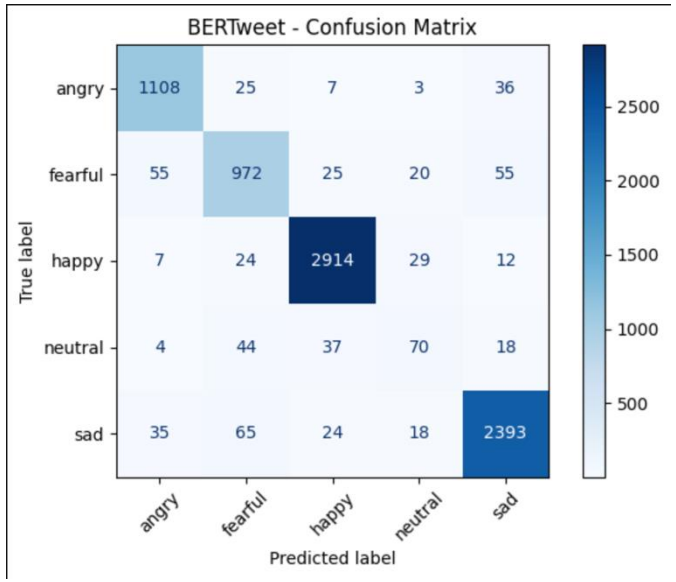
	precision	recall	f1-score	support
angry	0.87	0.89	0.88	491
fear	0.83	0.79	0.81	503
happy	0.85	0.81	0.83	503
neutral	0.83	0.90	0.86	517
sad	0.81	0.79	0.80	503
accuracy			0.83	2517
macro avg	0.83	0.83	0.83	2517
weighted avg	0.83	0.83	0.83	2517



Model 3 — Text Emotion Recognition (TER)

- ASR converts speech to text; BERTweet classifies the transcript
- Architecture: BERTweet (vinai/bertweet-base) fine-tuned via HuggingFace Transformers
- Pretrained on 850M tweets — natively handles slang, hashtags, and informal language
- Training: pashupatigupta + bhavikjikadara Kaggle datasets

	precision	recall	f1-score	support
angry	0.92	0.94	0.93	1179
fearful	0.86	0.86	0.86	1127
happy	0.97	0.98	0.97	2986
neutral	0.50	0.40	0.45	173
sad	0.95	0.94	0.95	2535
accuracy			0.93	8000
macro avg	0.84	0.83	0.83	8000
weighted avg	0.93	0.93	0.93	8000



Hyperparameter Tuning & Optimization

- **Method 1 — Architecture Comparison (SER)**
  - Tested BiLSTM+CNN vs. CNN+BiLSTM
  - BiLSTM+CNN: 89% accuracy, 94% F1 on Angry
  - CNN+BiLSTM: 75% accuracy, 73% F1 on Angry
  - Finding: BiLSTM needs direct access to sequential input — CNN after BiLSTM performs significantly better
- **Method 2 — Class Weight Balancing (all models)**
  - class_weight='balanced' applied across all three pipelines
  - Targets underrepresented classes (Angry, Fearful) without discarding data
  - Neutral class in FER required upweighting due to fewer samples
- **Method 3 — Transformer Fine-Tuning Schedule (TER)**
  - Warmup + decay learning rate schedule for BERTweet
  - Prevents catastrophic forgetting of pretrained representations
  - Batch size tuning (16 vs. 32) + early stopping on val_loss
- **Additional (FER):**
  - Investigated VGGNet and AlexNet architectures
  - Image augmentation during training to improve generalization to real-world lighting and pose

Live Demo — EmoTracker Dashboard

Questions?